

Basic Trigonometry – Ratios, Oblique Triangles, and Real-World Applications

Trigonometry is one of the most practical branches of mathematics. The word itself comes from the Greek words *trigonon* (triangle) and *metron* (measure), meaning "measurement of triangles." Every time a surveyor measures the height of a mountain, a pilot adjusts a flight path, a sailor sets a course, or an architect designs a roof, trigonometry is at work. It is the mathematics of angles and distances, and it allows us to calculate lengths and heights that cannot be measured directly with a ruler. Understanding trigonometry opens the door to careers in engineering, navigation, astronomy, architecture, and even video game design.

This reading material illustrates the trigonometric ratios in a right-angled triangle using real-life objects and figures. It explains how to find the area of oblique triangles using Heron's Formula and the formula $\text{Area} = \frac{1}{2}ab \sin C$. Finally, it demonstrates how to solve practical problems involving right and oblique triangles, angles of elevation and depression, and the use of bearings in navigation.

Part 1: The Trigonometric Ratios in a Right-Angled Triangle

What Is a Right-Angled Triangle?

A **right-angled triangle** is a triangle with one angle measuring exactly 90° . The side opposite the right angle is called the **hypotenuse** – it is always the longest side. The other two sides are called the **legs**.

Labeling the Sides

To use trigonometric ratios, we must label the sides relative to a chosen acute angle (an angle less than 90°). Let us call this angle θ (theta).

Side	Definition	Relative to Angle θ
Hypotenuse	The longest side, opposite the 90° angle	Always the same
Opposite	The side across from angle θ	Changes depending on which angle you choose
Adjacent	The side next to angle θ (not the hypotenuse)	Changes depending on which angle you choose

The Three Basic Trigonometric Ratios (SOH-CAH-TOA)

Ratio	Abbreviation	Formula	Memory Aid
Sine	$\sin \theta$	$\text{Opposite} \div \text{Hypotenuse}$	SOH
Cosine	$\cos \theta$	$\text{Adjacent} \div \text{Hypotenuse}$	CAH
Tangent	$\tan \theta$	$\text{Opposite} \div \text{Adjacent}$	TOA

SOH-CAH-TOA is the classic mnemonic: **S**ine = **O**pposite/**H**ypotenuse, **C**osine = **A**djacent/**H**ypotenuse, **T**angent = **O**pposite/**A**djacent.

Real-Life Illustration: The Flagpole

Imagine you are standing 12 meters away from the base of a flagpole. You look up at the top of the flagpole and measure the angle of elevation to be 35° . You want to know how tall the flagpole is – but you cannot climb it to measure.

Step 1: Draw and label the triangle.

- The flagpole is the **opposite** side (height, unknown).
- The distance from you to the base is the **adjacent** side (12 m).
- The angle of elevation is $\theta = 35^\circ$.

Step 2: Choose the correct ratio.

Since you know the adjacent side and want to find the opposite side, use **tangent**.

$$\tan \theta = \text{Opposite} \div \text{Adjacent}$$

Step 3: Substitute and solve.

$$\tan 35^\circ = h \div 12$$

$$h = 12 \times \tan 35^\circ$$

$$h = 12 \times 0.7002$$

$$h \approx 8.40 \text{ meters}$$

Answer: The flagpole is approximately **8.40 meters tall**.

The Special Angles

Certain angles appear so often in trigonometry that their ratios should be memorized.

Angle (θ)	$\sin \theta$	$\cos \theta$	$\tan \theta$
0°	0	1	0
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$

Angle (θ)	$\sin \theta$	$\cos \theta$	$\tan \theta$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	1	0	Undefined

The Reciprocal Ratios

Three additional ratios are derived from the basic three.

Ratio	Abbreviation	Formula
Cosecant	$\csc \theta$	Hypotenuse $\div$ Opposite (reciprocal of sine)
Secant	$\sec \theta$	Hypotenuse $\div$ Adjacent (reciprocal of cosine)
Cotangent	$\cot \theta$	Adjacent $\div$ Opposite (reciprocal of tangent)

Part 2: Solving Right Triangles

To **solve a right triangle** means to find all unknown sides and angles. You need at least one side and one acute angle, or two sides.

Example 1: Given One Angle and One Side

A ladder leans against a wall. The ladder is 5 meters long and makes a 65° angle with the ground. How high up the wall does the ladder reach?

Given: Hypotenuse = 5 m; angle with ground = 65°

Find: Height (opposite side)

$$\sin 65^\circ = \text{Opposite} \div \text{Hypotenuse}$$

$$\sin 65^\circ = h \div 5$$

$$h = 5 \times \sin 65^\circ$$

$$h = 5 \times 0.9063$$

$$h \approx 4.53 \text{ meters}$$

Answer: The ladder reaches approximately **4.53 meters** up the wall.

Example 2: Given Two Sides

A right triangle has a hypotenuse of 13 cm and one leg of 5 cm. Find the other leg and both acute angles.

Step 1: Find the missing leg using the Pythagorean Theorem.

$$a^2 + b^2 = c^2$$

$$5^2 + b^2 = 13^2$$

$$25 + b^2 = 169$$

$$b^2 = 144$$

$$b = 12 \text{ cm}$$

Step 2: Find one acute angle.

$$\sin \theta = \text{Opposite} \div \text{Hypotenuse} = 5 \div 13 \approx 0.3846$$

$$\theta = \sin^{-1}(0.3846) \approx 22.62^\circ$$

Step 3: Find the other angle.

Since the angles in a triangle sum to 180° , and one angle is 90° :

$$90^\circ + 22.62^\circ + \beta = 180^\circ$$

$$\beta = 67.38^\circ$$

Answer: The missing leg is **12 cm**, and the acute angles are **22.62°** and **67.38°**.

Part 3: Area of Oblique Triangles

An **oblique triangle** is any triangle that does not contain a right angle. It can be acute (all angles less than 90°) or obtuse (one angle greater than 90°). Two important formulas are used to find the area of oblique triangles.

Formula 1: Area = $\frac{1}{2} ab \sin C$

This formula is used when you know **two sides and the included angle**.

$$\text{Area} = \frac{1}{2} \times a \times b \times \sin C$$

Where:

- a and b are two sides of the triangle
- C is the angle between them (the included angle)

Example: Area Using $\frac{1}{2} ab \sin C$

A triangular garden has two sides measuring 8 m and 12 m, with an included angle of 55°. Find the area.

Step 1: Identify the given values.

$$a = 8 \text{ m}, b = 12 \text{ m}, C = 55^\circ$$

Step 2: Substitute into the formula.

$$\text{Area} = \frac{1}{2} \times 8 \times 12 \times \sin 55^\circ$$

$$\text{Area} = 48 \times 0.8192$$

$$\text{Area} \approx 39.32 \text{ m}^2$$

Answer: The area of the garden is approximately **39.32 square meters**.

Formula 2: Heron's Formula

Heron's Formula is used when you know **all three sides** of a triangle but no angles.

Step 1: Find the semi-perimeter (s).

$$s = (a + b + c) \div 2$$

Step 2: Apply Heron's Formula.

$$\text{Area} = \sqrt{s(s - a)(s - b)(s - c)}$$

Example: Area Using Heron's Formula

A triangular plot of land has sides measuring 7 m, 9 m, and 12 m. Find its area.

Step 1: Find the semi-perimeter.

$$s = (7 + 9 + 12) \div 2 = 28 \div 2 = 14 \text{ m}$$

Step 2: Apply Heron's Formula.

$$\text{Area} = \sqrt{14(14 - 7)(14 - 9)(14 - 12)}$$

$$\text{Area} = \sqrt{14 \times 7 \times 5 \times 2}$$

$$\text{Area} = \sqrt{980}$$

$$\text{Area} \approx 31.30 \text{ m}^2$$

Answer: The area of the land is approximately **31.30 square meters**.

Comparison of the Two Formulas

Formula	When to Use	What You Need to Know
$\text{Area} = \frac{1}{2} ab \sin C$	When you know two sides and the included angle	a, b, and angle C
Heron's Formula	When you know all three sides	a, b, and c

Part 4: The Law of Sines and the Law of Cosines

To solve oblique triangles, we use two additional laws.

The Law of Sines

$$a / \sin A = b / \sin B = c / \sin C$$

Use this when you know:

- Two angles and one side (AAS or ASA)
- Two sides and a non-included angle (SSA – the ambiguous case)

The Law of Cosines

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Use this when you know:

- Three sides (SSS)
- Two sides and the included angle (SAS)

Example: Solving an Oblique Triangle with the Law of Sines

In triangle ABC, angle A = 40°, angle B = 60°, and side a = 10 cm. Find side b.

$$a / \sin A = b / \sin B$$

$$10 / \sin 40^\circ = b / \sin 60^\circ$$

$$b = (10 \times \sin 60^\circ) \div \sin 40^\circ$$

$$b = (10 \times 0.8660) \div 0.6428$$

$$b \approx 13.47 \text{ cm}$$

Answer: Side b is approximately **13.47 cm**.

Example: Solving an Oblique Triangle with the Law of Cosines

In triangle ABC, $a = 5$ cm, $b = 7$ cm, and angle $C = 50^\circ$. Find side c .

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$c^2 = 25 + 49 - 2(5)(7)(\cos 50^\circ)$$

$$c^2 = 74 - 70(0.6428)$$

$$c^2 = 74 - 44.99$$

$$c^2 = 29.01$$

$$c \approx 5.39 \text{ cm}$$

Answer: Side c is approximately **5.39 cm**.

Part 5: Angles of Elevation and Depression

Definitions

Term	Definition
Angle of Elevation	The angle between the horizontal line of sight and the line of sight upward to an object
Angle of Depression	The angle between the horizontal line of sight and the line of sight downward to an object

Key Insight: The angle of elevation from point A to point B is equal to the angle of depression from point B to point A (alternate interior angles when horizontal lines are parallel).

Example 1: Angle of Elevation

From a point 50 meters from the base of a building, the angle of elevation to the top of the building is 62° . How tall is the building?

Given: Adjacent = 50 m; angle = 62°

Find: Opposite (height)

$$\tan 62^\circ = h \div 50$$

$$h = 50 \times \tan 62^\circ$$

$$h = 50 \times 1.8807$$

$$h \approx 94.04 \text{ meters}$$

Answer: The building is approximately **94.04 meters** tall.

Example 2: Angle of Depression

From the top of a 120-meter cliff, the angle of depression to a boat on the water is 28° . How far is the boat from the base of the cliff?

Given: Opposite (height) = 120 m; angle = 28°

Find: Adjacent (distance)

$$\tan 28^\circ = 120 \div d$$

$$d = 120 \div \tan 28^\circ$$

$$d = 120 \div 0.5317$$

$$d \approx 225.70 \text{ meters}$$

Answer: The boat is approximately **225.70 meters** from the base of the cliff.

Example 3: Two Angles of Elevation (Same Side)

From a point on the ground, the angle of elevation to the top of a tower is 30° . After walking 40 meters closer to the tower, the angle of elevation becomes 55° . Find the height of the tower.

Step 1: Let h = height of the tower and x = distance from the closer point to the base.

From the second position:

$$\tan 55^\circ = h \div x$$

$$h = x \tan 55^\circ$$

From the first position:

$$\tan 30^\circ = h \div (x + 40)$$

$$h = (x + 40) \tan 30^\circ$$

Step 2: Set the two expressions equal.

$$x \tan 55^\circ = (x + 40) \tan 30^\circ$$

$$1.4281x = 0.5774(x + 40)$$

$$1.4281x = 0.5774x + 23.096$$

$$0.8507x = 23.096$$

$$x \approx 27.15 \text{ m}$$

Step 3: Find h .

$$h = 27.15 \times \tan 55^\circ$$

$$h = 27.15 \times 1.4281$$

$$h \approx 38.78 \text{ meters}$$

Answer: The tower is approximately **38.78 meters tall**.

Part 6: Bearings in Navigation

What Is a Bearing?

A **bearing** is a direction measured clockwise from North. Bearings are always written with three digits (e.g., 045° , 120° , 270°).

Direction	Bearing
North	000°
East	090°
South	180°
West	270°
Northeast	045°
Southeast	135°
Southwest	225°
Northwest	315°

Example 1: Simple Bearing Problem

A ship sails 30 km on a bearing of 060°. How far north and how far east has it traveled?

Step 1: Draw and identify the angle from North.

The bearing 060° means 60° clockwise from North. The angle inside the triangle (with the North line) is 60°.

Step 2: Find the northward distance (adjacent to the angle).

$$\cos 60^\circ = \text{North} \div 30$$

$$\text{North} = 30 \times \cos 60^\circ = 30 \times 0.5 = 15 \text{ km}$$

Step 3: Find the eastward distance (opposite to the angle).

$$\sin 60^\circ = \text{East} \div 30$$

$$\text{East} = 30 \times \sin 60^\circ = 30 \times 0.8660 \approx 25.98 \text{ km}$$

Answer: The ship has traveled **15 km north** and approximately **25.98 km east**.

Example 2: Bearings and the Law of Cosines

A plane flies 200 km on a bearing of 070° , then turns and flies 150 km on a bearing of 160° . How far is the plane from its starting point?

Step 1: Find the included angle.

The bearing changes from 070° to 160° . The turn angle is:

$$160^\circ - 070^\circ = 90^\circ$$

So the included angle between the two flight paths is $180^\circ - 90^\circ = 90^\circ$? No – we need the angle at the turning point.

Careful analysis:

- First leg: bearing 070°
- Second leg: bearing 160°
- The angle between the two bearings (at the turning point) is $160^\circ - 070^\circ = 90^\circ$

Step 2: Use the Law of Cosines.

$$d^2 = 200^2 + 150^2 - 2(200)(150)(\cos 90^\circ)$$

$$d^2 = 40,000 + 22,500 - 0$$

$$d^2 = 62,500$$

$$d = 250 \text{ km}$$

Answer: The plane is **250 km** from its starting point.

Example 3: Bearings and the Law of Sines

Two ships leave the same port. Ship A sails 40 km on a bearing of 030° . Ship B sails 50 km on a bearing of 100° . How far apart are the two ships?

Step 1: Find the included angle at the port.

$$100^\circ - 30^\circ = 70^\circ$$

Step 2: Use the Law of Cosines.

$$d^2 = 40^2 + 50^2 - 2(40)(50)(\cos 70^\circ)$$

$$d^2 = 1,600 + 2,500 - 4,000(0.3420)$$

$$d^2 = 4,100 - 1,368$$

$$d^2 = 2,732$$

$$d \approx 52.27 \text{ km}$$

Answer: The two ships are approximately **52.27 km** apart.

Key Terms

Term	Definition
Trigonometry	The branch of mathematics dealing with the relationships between sides and angles of triangles
Right-angled triangle	A triangle containing one 90° angle
Hypotenuse	The longest side of a right triangle, opposite the right angle
Opposite side	The side across from a chosen acute angle
Adjacent side	The side next to a chosen acute angle (not the hypotenuse)
Sine (sin)	Ratio of opposite side to hypotenuse
Cosine (cos)	Ratio of adjacent side to hypotenuse
Tangent (tan)	Ratio of opposite side to adjacent side
SOH-CAH-TOA	Mnemonic for the three basic trigonometric ratios
Oblique triangle	A triangle with no right angle
Heron's Formula	Formula for the area of a triangle given three sides

Term	Definition
Law of Sines	$a/\sin A = b/\sin B = c/\sin C$
Law of Cosines	$c^2 = a^2 + b^2 - 2ab \cos C$
Angle of elevation	Angle measured upward from the horizontal
Angle of depression	Angle measured downward from the horizontal
Bearing	Direction measured clockwise from North, written with three digits

Review Questions

1. Define the three basic trigonometric ratios (sine, cosine, tangent) in terms of the sides of a right triangle.
2. A right triangle has an opposite side of 9 cm and an adjacent side of 12 cm. Find the hypotenuse, $\sin \theta$, $\cos \theta$, and $\tan \theta$.
3. A 6-meter ladder leans against a wall at a 70° angle with the ground. How high up the wall does the ladder reach?
4. Find the area of a triangle with sides 10 m and 14 m and an included angle of 42° . Use $\text{Area} = \frac{1}{2} ab \sin C$.
5. Find the area of a triangle with sides 8 cm, 11 cm, and 15 cm using Heron's Formula.
6. In triangle ABC, angle $A = 35^\circ$, angle $B = 75^\circ$, and side $a = 12$ cm. Use the Law of Sines to find side b .
7. In triangle ABC, $a = 6$ cm, $b = 9$ cm, and angle $C = 65^\circ$. Use the Law of Cosines to find side c .
8. From a point 80 meters from the base of a tower, the angle of elevation to the top is 48° . Find the height of the tower.
9. From the top of a 90-meter lighthouse, the angle of depression to a ship is 22° . How far is the ship from the base of the lighthouse?
10. Application question: A ship sails 60 km on a bearing of 045° , then turns and sails 80 km on a bearing of 135° .
 - (a) Find the angle between the two legs of the journey.
 - (b) Use the Law of Cosines to find the distance of the ship from its starting point.
 - (c) If the ship wants to return directly to port, what bearing should it take? (Hint: use the Law of Sines to find the angle, then convert to a bearing.)

Answer Key

1. $\sin \theta = O/H$, $\cos \theta = A/H$, $\tan \theta = O/A$

2. $c^2 = 9^2 + 12^2 = 225 \rightarrow \mathbf{c = 15 \text{ cm}}$

$\sin \theta = 9/15 = \mathbf{0.6}$; $\cos \theta = 12/15 = \mathbf{0.8}$; $\tan \theta = 9/12 = \mathbf{0.75}$

3. $\sin 70^\circ = h/6 \rightarrow h = 6 \times 0.9397 \rightarrow \mathbf{h \approx 5.64 \text{ m}}$

4. $\text{Area} = \frac{1}{2} \times 10 \times 14 \times \sin 42^\circ = 70 \times 0.6691 \rightarrow \mathbf{\text{Area} \approx 46.84 \text{ m}^2}$

5. $s = (8+11+15)/2 = 17$

$\text{Area} = \sqrt{[17 \times 9 \times 6 \times 2]} = \sqrt{1,836} \rightarrow \mathbf{\text{Area} \approx 42.85 \text{ cm}^2}$

6. $C = 180^\circ - 35^\circ - 75^\circ = 70^\circ$

$b = (12 \times \sin 75^\circ) / \sin 35^\circ = 11.5908 / 0.5736 \rightarrow \mathbf{b \approx 20.21 \text{ cm}}$

7. $c^2 = 6^2 + 9^2 - 2(6)(9)(\cos 65^\circ) = 117 - 45.64 = 71.36$

$c = \sqrt{71.36} \rightarrow \mathbf{c \approx 8.45 \text{ cm}}$

8. $\tan 48^\circ = h/80 \rightarrow h = 80 \times 1.1106 \rightarrow \mathbf{h \approx 88.85 \text{ m}}$

9. $\tan 22^\circ = 90/d \rightarrow d = 90 / 0.4040 \rightarrow \mathbf{d \approx 222.77 \text{ m}}$

10.

(a) $135^\circ - 045^\circ = \mathbf{90^\circ}$

(b) $d^2 = 60^2 + 80^2 - 2(60)(80)(\cos 90^\circ) = 10,000 \rightarrow \mathbf{d = 100 \text{ km}}$

(c) $\sin(\text{angle}) = 80/100 = 0.8 \rightarrow \text{angle} \approx 53.13^\circ$

Bearing to ship = $045^\circ + 53.13^\circ = 098.13^\circ$

Return bearing = $098.13^\circ + 180^\circ \rightarrow \mathbf{\approx 278^\circ}$